

New York University

CS-GY 6763: Homework 2.

Due Thursday, October 15th, 2026, 11:59pm ET.

Collaboration is allowed on this problem set, but solutions must be written-up individually. Please list collaborators for each problem separately, or write “No Collaborators” if you worked alone.

Problem 1: Exploring Concentration Bounds

(8 pts) Let X be a random variable uniformly distributed in the interval $[0, 1]$. Since we know X 's distribution exactly, we can easily check that $\Pr[X \geq 7/8] = 1/8$. But let's take a look at what various concentration inequalities would predict about this probability using less information about X .

1. Give an upper bound on $\Pr[X \geq 7/8]$ using Markov's inequality.
2. Give an upper bound on $\Pr[X \geq 7/8]$ using Chebyshev's inequality.
3. Give an upper bound on $\Pr[X \geq 7/8]$ by applying Markov's inequality to the random variable X^2 (the “raw” second moment). Note that this is slightly different than using Chebyshev's inequality, which applies Markov to the “central” second moment $(X - \mathbb{E}[X])^2$.
4. What happens for higher moments? Apply Markov's inequality to X^q for $q = 3, 4, \dots, 10$. Describe what you see in a sentence. What value of q gives the tightest bound?
5. One take-away here is that, depending on the random variable being studied, it is not always optimal to use the variance as a deviation measure to show concentration. Markov's inequality can be used with any monotonic function g , and as we see above, different choices might give better bounds. Exhibit a monotonic function g so that applying Markov's inequality to $g(X)$ gives as tight an upper bound on $\Pr[X \geq 7/8]$ as you can. Maximum points if you can get $\Pr[X \geq 7/8] \leq 1/8$, which would be the best possible.

Problem 2: Boosting Success Probability

(12 pts) We saw in class that computing the *mean* of many different answers from a randomized algorithm is often an effective way to improve accuracy. It turns out that the *median* is also very useful: it can be used to improve *success probability*.

In particular, suppose our goal is to estimate some scalar quantity $f(X)$ of a dataset X (e.g., the number of distinct items in a stream). Suppose we have developed a high-accuracy randomized algorithm, $\mathcal{A}$, but it only succeeds with probability $\geq 2/3$. I.e., on any input X , with probability $\geq 2/3$,

$$|\mathcal{A}[X] - f(X)| \leq \epsilon.$$

Prove that, if we run $\mathcal{A}$ a total of $r = O(\log(1/\delta))$ times with different random seeds and return the *median*, M , of the r outputs, then with probability $1 - \delta$:

$$|M - f(X)| \leq \epsilon.$$

This approach could be used, for example, to obtain an $O(\log(1/\delta)/\epsilon^2)$ space algorithm for distinct items, whereas in class we only proved $O(1/(\delta\epsilon^2))$ using Chebyshev's inequality.

Hint: Think about what event would have to happen for the median to be “bad”, i.e., to give error $> \epsilon$. If the median is bad, what does it say about the fraction of our r estimates that are bad?

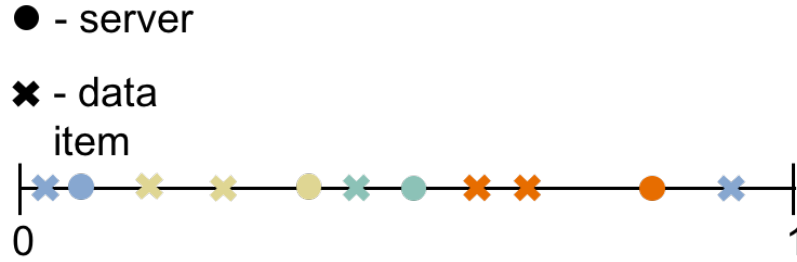


Figure 1: Each data item is stored on the server with matching color.

Problem 3: Hashing around the clock

(15 pts) In modern systems, hashing is often used to distribute data items or computational tasks to a collection of servers. What happens when a server is added or removed from a system? Most hash functions, including those discussed in class, are tailored to the number of servers, n , and would change completely if n changes. This would require rehashing and moving all of our m data items, an expensive operation.

Here we consider an approach to avoid this problem. Assume we have access to a completely random hash function that maps any value x to a real value $h(x) \in [0, 1]$. Use the hash function to map *both* data items and servers randomly to $[0, 1]$. Each data item is stored on the first server to its right on the number line (with wrap around – i.e. a job hashed below 1 but above all servers is assigned to the first server after 0). When a new server is added to the system, we hash it to $[0, 1]$ and move data items accordingly.

1. Suppose we have n servers initially. When a new server is added to the system, what is the expected number of data items that need to be relocated?
2. Show that, with probability $> 9/10$, no server “owns” more than an $O(\log n/n)$ fraction of the interval $[0, 1]$. **Hint:** This can be proven without a concentration bound.
3. Show that if we have n servers and m items and $m > n$, the maximum load on any server is no more than $O(\frac{m}{n} \log n)$ with probability $> 9/10$.

Problem 4: Johnson-Lindenstrauss Approximates Inner Products

(10 pts)

1. Suppose that Π is a Johnson-Lindenstrauss matrix with $k = O\left(\frac{\log(1/\delta)}{\epsilon^2}\right)$ rows. Prove that for any x, y :

$$|\langle x, y \rangle - \langle \Pi x, \Pi y \rangle| \leq \epsilon(\|x\|_2^2 + \|y\|_2^2)$$

with probability $\geq 1 - \delta$. **Hint:** Try exploiting connections between norms and inner products.

2. Show that a better bound can be obtained by augmenting our sketches with the norms of each vector. I.e., instead of just Πx and Πy , store as the sketches for x and y both $(\Pi x, \|x\|_2)$ and $(\Pi y, \|y\|_2)$ and show that using these sketches we can compute an estimate z such that

$$|\langle x, y \rangle - z| \leq \epsilon(\|x\|_2 \|y\|_2)$$

with $1 - \delta$ probability when $k = O\left(\frac{\log(1/\delta)}{\epsilon^2}\right)$. By the AM-GM inequality, we always have that $\|x\|_2 \|y\|_2 \leq \frac{\|x\|_2^2 + \|y\|_2^2}{2}$.

Problem 5: Analyzing Sign-JL and JL for Inner Products

(15 pts) Often practitioners prefer JL matrices with discrete random entries instead of Gaussians because they take less space to store and are easier to generate. We analyze one construction below.

Suppose that $\mathbf{\Pi}$ is a “sign Johnson-Lindenstrauss matrix” with n columns, k rows, and i.i.d. ± 1 entries scaled by $1/\sqrt{k}$. In other words, each entry in the matrix has values $-1/\sqrt{k}$ with probability $1/2$ and $1/\sqrt{k}$ with probability $1/2$.

1. Prove that for any vector $\mathbf{x} \in \mathbb{R}^n$, $\mathbb{E}[\|\mathbf{\Pi}\mathbf{x}\|_2^2] = \|\mathbf{x}\|_2^2$ and that $\text{Var}[\|\mathbf{\Pi}\mathbf{x}\|_2^2] \leq \frac{2}{k}\|\mathbf{x}\|_2^4$. This is the meat of the problem and will take some effort.

2. Use the above to prove that

$$\Pr[|\|\mathbf{\Pi}\mathbf{x}\|_2^2 - \|\mathbf{x}\|_2^2| \geq \epsilon\|\mathbf{x}\|_2^2] \leq \delta$$

as long as we choose $k = O(\frac{1}{\delta\epsilon^2})$. Note that this bound almost matches the distributed JL lemma proven in class, but with a worse failure probability dependence of $1/\delta$ in place of $\log(1/\delta)$.

With more work, it is possible to improve the dependence to $\log(1/\delta)$ for the sign-JL matrix, but we will not do so here.